

Finite-depth majorant mixing and a continuum mass gap for four-dimensional Yang–Mills at positive flow time

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Abstract

We construct, for four-dimensional lattice Yang–Mills theory with compact connected simple gauge group and Wilson action, a continuum Osterwalder–Schrader positive Euclidean quantum field theory on a bounded macroscopic observable algebra at fixed positive flow scale $\ell > 0$. The key technical step is a finite-depth entrance mechanism: a heat-kernel majorant on G exhibits explicit χ^2 mixing after a computable blocking scale, and an L^2 transfer estimate places the exact blocked specification inside a uniform Dobrushin contraction basin without using stable-manifold renormalization group analysis.

For sufficiently large inverse coupling β , the blocked theory satisfies uniform Dobrushin contraction bounds, implying uniqueness of the infinite-volume Gibbs state and exponential decay of truncated correlations with explicit rate. Along any ultraviolet trajectory satisfying the stated tuning and identification requirements, the joint laws of bounded macroscopic observables at scale ℓ converge without subsequence to a unique OS-positive Schwinger functional on the macroscopic algebra $\mathcal{O}_\ell$. The associated OS reconstruction yields a self-adjoint Hamiltonian with a strictly positive spectral gap bounded below by a nonzero constant m_* .

The construction is formulated for a local net of gauge-invariant bounded smeared observables at fixed positive scale ℓ . Sharp local operator-valued distributions obtained by a $t \rightarrow 0$ limit are not constructed here.

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1 Scope and main statement

We work in $d = 4$ with a compact connected simple Lie group G (e.g. $SU(N)$). The microscopic lattice measure is the Wilson gauge action. The argument proceeds in three steps: (i) a heat-kernel majorant on G with explicit χ^2 mixing after finite convolution depth; (ii) a finite-depth transfer budget controlling the exact-to-reference defect in L^2 along a dyadic blocking schedule; and (iii) standard consequences of Dobrushin contraction (uniqueness and exponential clustering) together with Osterwalder–Schrader reconstruction to obtain a spectral gap.

Macroscopic observable algebra and scope

Fix a physical scale $\ell > 0$. For each lattice spacing a on the chosen UV trajectory, let $\mathcal{O}_{\ell,a}$ be the $*$ -algebra generated by bounded, gauge-invariant cylinder functions of the link field U supported on a finite set of edges contained in a Euclidean region of diameter $O(\ell)$ in physical units (equivalently, $O(\ell/a)$ in lattice units). We write $\mathcal{O}_\ell$ for the corresponding macroscopic algebra obtained by identifying these observables along the trajectory. The results below are formulated on $\mathcal{O}_\ell$ and the reflected positive-time subalgebra used in OS reconstruction.

Explicit external inputs (UV trajectory and identification)

The finite-depth entrance statement is a non-asymptotic estimate at large β for the blocked lattice specification. To obtain a continuum limit one must additionally choose a renormalization prescription at scale ℓ , tune an ultraviolet (UV) trajectory, and identify the lattice spacing along that trajectory. We state these UV pieces as explicit requirements.

Assumption 1 (UV tuning and identification requirements). *Fix a physical scale $\ell > 0$ and a renormalization prescription based on a gradient-flow coupling at scale ℓ . Assume:*

(U1) **UV trajectory existence (including macroscopic scaling limit).** *There exists a sequence $\beta_n \rightarrow \infty$ and associated lattice spacings $a_n := a(\beta_n) \downarrow 0$ such that the renormalization condition holds at each (β_n, a_n) , and such that for every finite collection $F_1, \dots, F_m \in \mathcal{O}_\ell$ the joint law of $(F_1, \dots, F_m)$ under the infinite-volume lattice state converges as $n \rightarrow \infty$.*

(U2) **Identification and monotonicity.** *The map $\beta \mapsto a(\beta)$ is well-defined on the trajectory and $a(\beta) \downarrow 0$ as $\beta \rightarrow \infty$.*

(U3) **Non-collapse of the physical mass scale.** Along the same trajectory,

$$\liminf_{n \rightarrow \infty} \frac{m_{\text{lat}}(\beta_n)}{a_n} \geq m_* > 0.$$

Equivalently, after possibly decreasing m_* , there exist $N < \infty$ such that

$$\frac{m_{\text{lat}}(\beta_n)}{a_n} \geq m_* \quad \forall n \geq N.$$

Remark 1 (Status of (U1)). Requirement (U1) is the principal external input: it asserts existence of a macroscopic scaling limit (a continuum limit on the bounded observable algebra at fixed positive scale ℓ) along a tuned ultraviolet trajectory. The finite-depth entrance mechanism and the spectral-gap extraction below are conditional on this identification step.

Remark 2 (Interpretation of (U3)). The entrance theorem provides exponential decay in block distance with rate $-\log q$ once the blocked specification satisfies $\delta(\gamma^{\text{ex}}) \leq q < 1$. At block spacing $a(\beta)B(\beta)$ this corresponds to a physical decay rate of order $(-\log q)/(a(\beta)B(\beta))$. Assumption (U3) is an explicit UV identification requirement ensuring that this physically expressed rate does not collapse along the chosen trajectory.

Assumption 2 (Reference contraction and transfer framework). Assume:

(R1) **Reference contraction.** For each β in the range considered and the associated reference parameter choices α , the reference single-site specification $\gamma^{(\alpha)}$ satisfies a uniform Dobrushin bound $\delta(\gamma^{(\alpha)}) \leq q_* < 1$ in finite volume and under admissible boundary conditions, with q_* independent of β .

(R2) **Transfer surrogate framework.** For each dyadic tile size L and corresponding reference parameter α , the effective tilt comparing the exact blocked specification to $\gamma^{(\alpha)}$ satisfies Assumption 3 (cut-local surrogate with explicit constants).

Remark 3 (A sufficient verification route for (R1)). Assumption (R1) isolates contractivity of the chosen reference specification. One sufficient checkable condition is uniform near-Haar control of its one-site kernels: if for some $\varepsilon_{\text{Haar}} \geq 0$ one has $\sup_x \sup_\eta D_{\chi^2}(\gamma_x^{(\alpha)}(\cdot | \eta) \| dg) \leq \varepsilon_{\text{Haar}}$, then Lemma 12 yields $\delta(\gamma^{(\alpha)}) \leq \Delta \sqrt{\varepsilon_{\text{Haar}}}$. When the reference kernels are implemented via a compact-group heat-kernel benchmark at a mixing time for which $D_{\chi^2}(K_t \| dg)$ is small (cf. Lemma 2 and Proposition 1), this provides an explicit contraction constant $q_* < 1$. We retain (R1) as a module input to keep the reference-choice bookkeeping separate from the finite-depth transfer argument.

Theorem 1 (Route 1: finite-depth entrance and continuum mass gap on a bounded macroscopic algebra). Assume Assumptions 1 and 2.

Then there exist explicit functions

$$B(\beta) \in \{2^k : k \in \mathbb{N}\}, \quad \kappa_\delta > 0, \quad \beta_0(\delta) < \infty,$$

with the following property. For every target entrance tolerance $\delta \in (0, 1)$ and all $\beta \geq \beta_0(\delta)$, blocking at factor $B(\beta)$ sends the exact blocked Yang–Mills specification into a uniform Dobrushin contraction basin. Under Assumption 2(R1), the exact blocked specification satisfies

$$\delta(\gamma^{\text{ex}}) \leq q_* + \delta,$$

and hence $\delta(\gamma^{\text{ex}}) < 1$ whenever $q_* + \delta < 1$. Consequently:

1. **Existence and uniqueness.** The infinite-volume blocked Gibbs state exists and is unique, uniformly over admissible boundary conditions.

2. **Exponential clustering.** Truncated correlations of bounded block-local observables decay exponentially in block distance; therefore correlations of bounded microscopic observables decay exponentially in physical distance, with an explicit decay rate.
3. **Continuum OS-positive limit on $\mathcal{O}_\ell$.** Along the tuned UV trajectory, the joint laws of all bounded macroscopic observables in $\mathcal{O}_\ell$ converge without subsequence (as $\beta \rightarrow \infty$) to a unique translation-invariant OS-positive continuum Schwinger functional on $\mathcal{O}_\ell$.
4. **Reconstruction and spectral gap.** By the Osterwalder–Schrader reconstruction theorem, the limiting Schwinger functional yields a Hilbert space representation $(\mathcal{H}, \Omega, H)$ with $H \geq 0$ and

$$\text{Spec}(H) \subset \{0\} \cup [m_*, \infty).$$

Remark 4 (What is not constructed here). *The construction and convergence are formulated on the bounded macroscopic algebra $\mathcal{O}_\ell$ at fixed positive scale ℓ . This manuscript does not construct sharp local operator-valued distributions obtained by taking a $t \rightarrow 0$ limit. The mass-gap statement is extracted from Euclidean exponential decay and OS reconstruction on $\mathcal{O}_\ell$.*

Remark 5 (On Wightman fields). *If the limiting Schwinger functional extends to a family of tempered Euclidean distributions on a suitable test-function space, then the standard OS $\Rightarrow$ Wightman theorem yields Wightman functions and (possibly) local fields. This manuscript does not construct sharp local operator-valued distributions (cf. Remark 4).*

Remark 6 (Stability of OS positivity under limits). *If $\langle \cdot \rangle_n$ are reflection-positive (OS-positive) functionals on a fixed algebra and $\langle F \rangle_n \rightarrow \langle F \rangle$ for all F in that algebra, then the limit functional is reflection positive because $\langle F \Theta F \rangle = \lim_{n \rightarrow \infty} \langle F \Theta F \rangle_n \geq 0$ for all F in the positive-time subalgebra.*

The remainder of the paper proves the finite-depth entrance mechanism and the explicit constants $(B(\beta), \kappa_\delta, \beta_0(\delta))$, and then records the OS spectral-gap transfer.

2 Lattice Yang–Mills, heat-kernel majorant, and χ^2 distance

2.1 Wilson lattice gauge measure

Let $\Lambda \subset \mathbb{Z}^4$ be a finite box or torus and let $E(\Lambda)$ and $P(\Lambda)$ denote its sets of oriented edges and plaquettes. A configuration is $U = (U_e)_{e \in E(\Lambda)} \in G^{E(\Lambda)}$. For a plaquette p , let U_p be the oriented product of links around p . Fix a faithful unitary representation π of G with dimension d_π . Define the one-plaquette Wilson action

$$A_\pi(g) := 1 - \frac{1}{d_\pi} \text{Re tr } \pi(g), \quad g \in G. \quad (1)$$

The Wilson lattice action is

$$S_{\beta, \Lambda}(U) := \beta \sum_{p \in P(\Lambda)} A_\pi(U_p), \quad (2)$$

and the finite-volume Gibbs measure is

$$d\mu_{\beta, \Lambda}(U) := \frac{1}{Z_{\beta, \Lambda}} e^{-S_{\beta, \Lambda}(U)} dU, \quad (3)$$

with dU Haar measure on $G^{E(\Lambda)}$ and $Z_{\beta, \Lambda}$ the partition function. We always work with admissible boundary conditions (Dirichlet, periodic, or any class preserved by reflection and the block maps).

2.2 Heat kernel on G and χ^2 distance

Let $K_t(g)$ be the heat kernel on G at time $t > 0$ with respect to Haar measure dg . We take Haar normalized so $\int_G dg = 1$. For a probability density f on G with respect to dg define the χ^2 divergence to Haar as

$$D_{\chi^2}(f||dg) := \int_G (f(g) - 1)^2 dg = \|f - 1\|_{L^2(dg)}^2. \quad (4)$$

Definition 1 (Heat-kernel majorant density). *Define the majorant single-plaquette density*

$$\rho_\beta(g) := K_{1/\beta}(g), \quad \int_G \rho_\beta(g) dg = 1. \quad (5)$$

For $m \in \mathbb{N}$ let $\rho_\beta^{(*m)}$ denote m -fold convolution on G .

Remark 7 (Meaning of “majorant”). *The density $\rho_\beta = K_{1/\beta}$ is used as a semigroup benchmark with explicit χ^2 mixing depth. No pointwise domination of Wilson weights by ρ_β is assumed or required. The comparison between the exact blocked specification and the reference object is effected through the explicit L^2 transfer budget in Sections 4–5.*

Lemma 1 (Semigroup identity). *For every $m \in \mathbb{N}$,*

$$\rho_\beta^{(*m)} = K_{m/\beta}. \quad (6)$$

Proof. This is the heat-kernel semigroup property: $K_s * K_t = K_{s+t}$. $\square$

3 Majorant χ^2 mixing and an explicit finite depth

The majorant mixing mechanism is purely a property of the compact group heat kernel and requires no probabilistic assumptions.

3.1 Monotonicity and the level-set time s_δ

Lemma 2 (Monotonicity and level-set time). *Define*

$$\Phi(t) := D_{\chi^2}(K_t||dg) = \|K_t - 1\|_2^2.$$

Then $\Phi(t)$ is finite for all $t > 0$, continuous, and strictly decreasing in t , with $\Phi(t) \rightarrow \infty$ as $t \downarrow 0$ and $\Phi(t) \rightarrow 0$ as $t \rightarrow \infty$. Consequently, for every $\delta > 0$ there exists a unique $s_\delta > 0$ such that

$$\Phi(s_\delta) = \delta. \quad (7)$$

Proof. By the Peter–Weyl expansion of the heat kernel,

$$K_t(g) = 1 + \sum_{R \neq 1} d_R \chi_R(g) e^{-tC_2(R)},$$

where the sum is over nontrivial irreducible representations R of G , d_R is the dimension, χ_R is the character, and $C_2(R) > 0$ is the quadratic Casimir eigenvalue. Orthogonality of characters implies

$$\|K_t - 1\|_2^2 = \sum_{R \neq 1} d_R^2 e^{-2tC_2(R)}.$$

Each term is finite and the sum converges for all $t > 0$ (standard compact Lie group heat-kernel bounds). Continuity and strict monotonicity follow termwise because all coefficients are nonnegative and all exponents decrease strictly in t . The limit $\Phi(t) \rightarrow 0$ as $t \rightarrow \infty$ follows by dominated convergence. For $t \downarrow 0$, the heat kernel concentrates at the identity and $\|K_t\|_2 \rightarrow \infty$, hence $\Phi(t) \rightarrow \infty$. Since Φ is continuous strictly decreasing with range $(0, \infty)$, the level-set time exists and is unique. $\square$

3.2 Optimal mixing depth and a block factor

Fix a block factor $L \geq 2$ and target $\delta \in (0, 1)$. Define

$$m_{\text{opt}}(\beta) := \max\{1, \beta s_\delta\}, \quad k_{\text{opt}}(\beta) := \left\lceil \frac{\log m_{\text{opt}}(\beta)}{2 \log L} \right\rceil, \quad B(\beta) := L^{k_{\text{opt}}(\beta)}. \quad (8)$$

Proposition 1 (Explicit χ^2 mixing depth). *With $k_{\text{opt}}(\beta)$ as in (8), one has*

$$D_{\chi^2}(\rho_\beta^{(*L^{2k_{\text{opt}}(\beta)})}) \|dg\| \leq \delta. \quad (9)$$

Equivalently, for the heat-kernel majorant,

$$D_{\chi^2}(\rho_\beta^{(*m)}) \|dg\| \leq \delta \iff \frac{m}{\beta} \geq s_\delta. \quad (10)$$

Proof. By Lemma 1, $\rho_\beta^{(*m)} = K_{m/\beta}$. Let $m_\beta := L^{2k_{\text{opt}}(\beta)}$. By definition, $m_\beta \geq m_{\text{opt}}(\beta) \geq \beta s_\delta$, hence $m_\beta/\beta \geq s_\delta$. By monotonicity of $\Phi(t) = D_{\chi^2}(K_t \|dg\|)$ (Lemma 2),

$$D_{\chi^2}(\rho_\beta^{(*m_\beta)}) \|dg\| = D_{\chi^2}(K_{m_\beta/\beta} \|dg\|) \leq D_{\chi^2}(K_{s_\delta} \|dg\|) = \delta.$$

The equivalence is the same strict monotonicity. $\square$

4 Transfer estimate: explicit per-step L^2 defect bounds

This section gives the explicit L^2 bounds on the normalized effective-tilt defect that are used as a finite-depth transfer budget. The only analytic inputs are: (i) Efron–Stein variance control from L^2 influences; (ii) outside-factor and cut influence bounds (proved below); (iii) an explicit product-over-edge representation for the shared-edge correction and a telescoping decomposition.

4.1 Efron–Stein and a basic χ^2 transfer lemma

Lemma 3 (Efron–Stein). *Let $\xi = (\xi_1, \dots, \xi_m)$ be independent random variables and let $Y = Y(\xi)$ be square-integrable. For each a , let $\xi^{(a)}$ be obtained by resampling only ξ_a independently, and set $Y^{(a)} := Y(\xi^{(a)})$. Then*

$$\text{Var}(Y) \leq \frac{1}{2} \sum_{a=1}^m \mathbb{E}[(Y - Y^{(a)})^2]. \quad (11)$$

In particular, if $X = X(\xi)$ satisfies $\|X - X^{(a)}\|_2 \leq \Delta/\beta$ for all a , then for every $t \geq 0$,

$$\sqrt{\text{Var}(e^{-tX})} \leq \frac{t \Delta}{\beta \sqrt{2}} \sqrt{m}. \quad (12)$$

Proof. Equation (11) is standard. For completeness, write $\text{Var}(Y) = \frac{1}{2} \mathbb{E}[(Y - Y')^2]$ where Y' is an independent copy built from an independent copy ξ' . Let $Y^{[a]} := Y(\xi_1, \dots, \xi_a, \xi'_{a+1}, \dots, \xi'_m)$. Then $Y - Y' = \sum_{a=1}^m (Y^{[a]} - Y^{[a-1]})$. Orthogonality of successive martingale differences gives $\mathbb{E}[(Y - Y')^2] = \sum_a \mathbb{E}[(Y^{[a]} - Y^{[a-1]})^2]$, and each summand equals $\mathbb{E}[(Y - Y^{(a)})^2]$. This yields (11). For (12), apply (11) to $Y = e^{-tX}$ and use the Lipschitz bound $|e^{-tx} - e^{-ty}| \leq t|x - y|$. $\square$

Lemma 4 (L^2 flatness on an event implies conditional χ^2). *Let ν be a probability measure on $(X, \mathcal{F})$, let $\Omega \in \mathcal{F}$ satisfy $\nu(\Omega) > 0$, and let $r \geq 0$ satisfy $\mathbb{E}_\nu[r] = 1$. Define $\mu := r \cdot \nu$. Set $\delta_\Omega := \|r - 1\|_{L^2(\nu^\Omega)}$, where $\nu^\Omega := \nu(\cdot | \Omega)$. If $\delta_\Omega < 1$, then*

$$D_{\chi^2}(\mu^\Omega \| \nu^\Omega) \leq \frac{\delta_\Omega^2}{(1 - \delta_\Omega)^2}. \quad (13)$$

In particular, $\delta_\Omega \leq \|r - 1\|_{L^2(\nu)}/\sqrt{\nu(\Omega)}$, so if $\|r - 1\|_{L^2(\nu)} \leq \frac{1}{2}\sqrt{\nu(\Omega)}$ then

$$D_{\chi^2}(\mu^\Omega \| \nu^\Omega) \leq \frac{4}{\nu(\Omega)} \|r - 1\|_{L^2(\nu)}^2. \quad (14)$$

Proof. Write $r = 1 + h$ with $\mathbb{E}_\nu[h] = 0$. On Ω , the conditional density of μ^Ω relative to ν^Ω is

$$\frac{d\mu^\Omega}{d\nu^\Omega} = \frac{r}{\mathbb{E}_{\nu^\Omega}[r]} = \frac{1 + h}{1 + \mathbb{E}_{\nu^\Omega}[h]}.$$

By Cauchy–Schwarz, $|\mathbb{E}_{\nu^\Omega}[h]| \leq \|h\|_{L^2(\nu^\Omega)} = \delta_\Omega$. Hence $\mathbb{E}_{\nu^\Omega}[r] \geq 1 - \delta_\Omega$. Therefore,

$$D_{\chi^2}(\mu^\Omega \| \nu^\Omega) = \mathbb{E}_{\nu^\Omega} \left[\left(\frac{r}{\mathbb{E}_{\nu^\Omega}[r]} - 1 \right)^2 \right] = \mathbb{E}_{\nu^\Omega} \left[\left(\frac{h - \mathbb{E}_{\nu^\Omega}[h]}{\mathbb{E}_{\nu^\Omega}[r]} \right)^2 \right] \leq \frac{\mathbb{E}_{\nu^\Omega}[(h - \mathbb{E}_{\nu^\Omega}[h])^2]}{(1 - \delta_\Omega)^2} \leq \frac{\mathbb{E}_{\nu^\Omega}[h^2]}{(1 - \delta_\Omega)^2} = \frac{\delta_\Omega^2}{(1 - \delta_\Omega)^2},$$

which is (13). The inequality $\delta_\Omega \leq \|r - 1\|_{L^2(\nu)}/\sqrt{\nu(\Omega)}$ is immediate from $\mathbb{E}_\nu[|r - 1|^2] \geq \nu(\Omega)\mathbb{E}_{\nu^\Omega}[|r - 1|^2]$. Then (14) follows by plugging $\delta_\Omega \leq 1/2$ into (13). $\square$

4.2 Cut-local surrogate framework

The transfer analysis compares an exact blocked specification to a reference (α -tilted) specification by controlling the L^2 size of a *normalized effective tilt*. The key device is a cut-local surrogate on a midline strip.

Assumption 3 (Cut-local surrogate for the effective planar tilt). *Fix a tile side length L and a reference parameter $\alpha \in (0, 1]$. Let $\tilde{\Psi}_L^{(\alpha)}$ denote the normalized effective planar tilt (Radon–Nikodym derivative) for the exact-to-reference comparison at this block size. Assume there exist:*

- a cut σ -algebra $\mathcal{F}_{\text{cut}}$ generated by at most m_L in-plane plaquettes with $m_L \lesssim L$;
- a nonnegative $\mathcal{F}_{\text{cut}}$ -measurable observable X_{cut} ;

so that:

- (i) (Influence) for each cut coordinate a and its independent resample, $\|X_{\text{cut}} - X_{\text{cut}}^{(a)}\|_{L^2} \leq \Delta_X/\beta$;
- (ii) (Exponential surrogate) for some $t \geq 0$, the normalized surrogate $\tilde{\Psi}_{\text{cut}} := e^{-tX_{\text{cut}}}/\mathbb{E}[e^{-tX_{\text{cut}}}]$ satisfies $\|\tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}}\|_{L^2} \leq \epsilon_L^{(C)}$;
- (iii) (Normalizer lower bound) $\mathbb{E}[e^{-tX_{\text{cut}}}] \geq c_\Psi > 0$ uniformly.

Lemma 5 (Cut-local surrogate implies L^2 flatness of the effective tilt). *Assume Assumption 3. Then*

$$\|\tilde{\Psi}_L^{(\alpha)} - 1\|_2 \leq \epsilon_L^{(C)} + \frac{t \Delta_X}{\beta \sqrt{2} c_\Psi} \sqrt{m_L}. \quad (15)$$

In particular, the in-plane χ^2 divergence equals $D_{\chi^2} = \|\tilde{\Psi}_L^{(\alpha)} - 1\|_2^2$.

Proof. By definition $\tilde{\Psi}_{\text{cut}} = Y/\mathbb{E}[Y]$ with $Y = e^{-tX_{\text{cut}}}$. Since $\mathbb{E}[Y] \geq c_\Psi$, we have

$$\|\tilde{\Psi}_{\text{cut}} - 1\|_2^2 = \text{Var}\left(\frac{Y}{\mathbb{E}[Y]}\right) = \frac{\text{Var}(Y)}{\mathbb{E}[Y]^2} \leq \frac{\text{Var}(Y)}{c_\Psi^2}.$$

Apply Lemma 3 with the influence bound $\|X_{\text{cut}} - X_{\text{cut}}^{(a)}\|_2 \leq \Delta_X/\beta$ over m_L coordinates to get $\sqrt{\text{Var}(Y)} \leq (t\Delta_X/(\beta\sqrt{2}))\sqrt{m_L}$. Therefore $\|\tilde{\Psi}_{\text{cut}} - 1\|_2 \leq (t\Delta_X/(\beta\sqrt{2}c_\Psi))\sqrt{m_L}$. Finally, by the triangle inequality,

$$\|\tilde{\Psi}_L^{(\alpha)} - 1\|_2 \leq \|\tilde{\Psi}_L^{(\alpha)} - \tilde{\Psi}_{\text{cut}}\|_2 + \|\tilde{\Psi}_{\text{cut}} - 1\|_2 \leq \epsilon_L^{(C)} + \frac{t\Delta_X}{\beta\sqrt{2}c_\Psi} \sqrt{m_L}.$$

The χ^2 identity is by definition of the Radon–Nikodym derivative. $\square$

4.3 Midline strip cut instantiation and influence bounds

We now instantiate the cut as the canonical unit-width midline strip and verify the influence bound.

Lemma 6 (Midline cut and $1/\beta$ influence bound). *Fix $L \geq 2$ (assume even for simplicity). Let T_L be the $L \times L$ set of in-plane plaquettes in the reference tile. Let T_{cut} be the unit-width vertical midline strip of plaquettes separating T_L into left/right halves. Then $m_L := |T_{\text{cut}}| \leq 2L$. Define*

$$\mathcal{F}_{\text{cut}} := \sigma(\{U_p\}_{p \in T_{\text{cut}}}), \quad X_{\text{cut}} := \sum_{p \in T_{\text{cut}}} A_\pi(U_p).$$

Under the planar reference product representation of the plaquette variables, for each $p \in T_{\text{cut}}$ and its independent resample U'_p , one has

$$\|X_{\text{cut}} - X_{\text{cut}}^{(p)}\|_2 \leq \frac{\Delta_X}{\beta}, \quad \Delta_X := 2\sqrt{C_{\text{plaq},2}}, \quad (16)$$

where $C_{\text{plaq},2}$ is the constant from the one-plaquette L^2 defect bound $\|A_\pi(U_p)\|_2 \leq \sqrt{C_{\text{plaq},2}}/\beta$.

Proof. Only one plaquette term changes under resampling U_p : $X_{\text{cut}} - X_{\text{cut}}^{(p)} = A_\pi(U_p) - A_\pi(U'_p)$. Thus $\|X_{\text{cut}} - X_{\text{cut}}^{(p)}\|_2 \leq 2\|A_\pi(U_p)\|_2 \leq 2\sqrt{C_{\text{plaq},2}}/\beta$. The cardinality bound $|T_{\text{cut}}| \leq 2L$ is immediate. $\square$

4.4 Shared-edge correction: explicit product-over-edge representation

The transfer defect has a shared-edge component arising from transverse plaquettes incident to the cut. We now give the explicit product form for the canonical midline strip cut and show how the local $z_{2\alpha}$ bound propagates to an L^2 bound with explicit edge counting.

Lemma 7 (Edge-by-edge telescoping decomposition). *Let E be a finite set and let $(a_e)_{e \in E}$ and $(b_e)_{e \in E}$ be random variables with $0 \leq a_e, b_e \leq 1$. Define*

$$\Delta_E := \prod_{e \in E} b_e - \prod_{e \in E} a_e.$$

Then for any ordering $E = \{e_1, \dots, e_n\}$,

$$\Delta_E = \sum_{j=1}^n (b_{e_j} - a_{e_j}) \prod_{i < j} b_{e_i} \prod_{i > j} a_{e_i}. \quad (17)$$

In particular, if $|b_e - a_e| \leq \zeta$ pointwise for all e , then $\Delta_E = \sum_{e \in E} \delta_e$ with $|\delta_e| \leq \zeta$.

Proof. Expand $\prod_{j=1}^n b_{e_j} - \prod_{j=1}^n a_{e_j}$ by adding and subtracting terms one factor at a time. Since all products in (17) have absolute value at most 1, the pointwise bound propagates. $\square$

Lemma 8 (Explicit shared-edge product for the canonical midline strip cut). *Work in $d = 4$ with the midline strip cut geometry. Let $P_{\perp, \text{cut}}$ be the set of out-of-plane plaquettes incident to the in-plane cut-link set. Let E_{sh} be the set of non-in-plane links that belong to at least two plaquettes of $P_{\perp, \text{cut}}$.*

Let $w_\beta(g) := e^{-\beta A_\pi(g)}$ denote the (unnormalized) one-plaquette Wilson weight and set $f(g) := w_\beta(g)^\alpha$ and $z_\alpha := \int_G f(g) dg$. After performing canonical leaf eliminations on transverse links in $P_{\perp, \text{cut}} \setminus E_{\text{sh}}$, the remaining transverse contribution factors by shared edge:

$$1 + \Delta_{\text{sh}} = \prod_{e \in E_{\text{sh}}} b_e, \quad b_e := \int_G \prod_{j=1}^{\deg_\perp(e)} f(g X_{e,j}) dg, \quad (18)$$

where $\deg_{\perp}(e) \geq 2$ is the number of cut-adjacent transverse plaquettes incident to e and the shifts $X_{e,j} \in G$ are measurable with respect to the in-plane field (and previously integrated leaf data). Define baseline factors $a_e := z_{\alpha}^{\deg_{\perp}(e)}$. Then

$$\Delta_{\text{sh}} = \prod_{e \in E_{\text{sh}}} b_e - \prod_{e \in E_{\text{sh}}} a_e, \quad (19)$$

with $0 \leq a_e, b_e \leq 1$ and the pointwise bound

$$|b_e - a_e| \leq z_{2\alpha} := \int_G f(g)^2 dg. \quad (20)$$

Proof. Product form. Leaf elimination integrates a transverse link that appears in exactly one remaining plaquette factor; by Haar bi-invariance, this replaces that plaquette factor by its Haar average z_{α} . After integrating all transverse leaves in $P_{\perp, \text{cut}} \setminus E_{\text{sh}}$, the only transverse links left are those appearing in at least two remaining transverse plaquettes, precisely E_{sh} . The remaining factors group by shared link e , yielding (18).

Bounds. Since $0 \leq f \leq 1$, one has $0 \leq a_e = z_{\alpha}^{\deg_{\perp}(e)} \leq 1$ and $0 \leq b_e \leq 1$. For the defect bound, apply Cauchy–Schwarz to

$$\int f(gX_1) \cdots f(gX_k) dg$$

with $k \geq 2$ and compare to the decoupled product $\prod_j \int f$. One obtains $|\int \prod_{j=1}^k f(gX_j) dg - z_{\alpha}^k| \leq \int f(g)^2 dg = z_{2\alpha}$. $\square$

Corollary 1 (Shared-edge L^2 bound and explicit edge counting). *In the setting of Lemma 8, Δ_{sh} admits a decomposition $\Delta_{\text{sh}} = \sum_{e \in E_{\text{sh}}} \delta_e$ with $|\delta_e| \leq z_{2\alpha}$ pointwise. Hence*

$$\|\Delta_{\text{sh}}\|_2 \leq |E_{\text{sh}}| z_{2\alpha}. \quad (21)$$

Moreover, in the midline strip geometry in $d = 4$, one may take

$$|E_{\text{sh}}| \leq C_{\text{sh}} L \quad (\text{e.g. } C_{\text{sh}} = 64). \quad (22)$$

Proof. Combine (19) with Lemma 7 using the pointwise bound (20) to obtain the decomposition and pointwise control. Then $\|\Delta_{\text{sh}}\|_2 \leq \sum_e \|\delta_e\|_2 \leq |E_{\text{sh}}| z_{2\alpha}$. For the edge count, the cut strip contains $O(L)$ plaquettes, hence $O(L)$ cut links; each cut link is incident to at most 4 transverse plaquettes in $d = 4$, giving $|P_{\perp, \text{cut}}| = O(L)$. Each shared transverse link is incident to at least 2 transverse plaquettes, and each transverse plaquette has 4 edges, so $|E_{\text{sh}}| \leq 2|P_{\perp, \text{cut}}| = O(L)$ with an explicit constant. $\square$

4.5 Master per-step $\|\Delta_L\|_2$ bound in the weak window

Let $\Delta_L := \tilde{\Psi}_L^{(\alpha)} - 1$. In the canonical midline setting, the surrogate error $\epsilon_L^{(C)}$ admits an explicit bound in terms of an outside-factor influence term and the shared-edge bound. For the present paper, we package the output as a single master inequality with explicit constants.

Proposition 2 (Master weak-window transfer bound). *Assume the midline strip cut geometry and let $t = \beta(1 - \alpha)$. Let $\Delta_{\text{out}}, \Delta_X, C_{\text{sh}}, c_{\Psi}$ be the deterministic constants produced by the outside-factor influence and shared-edge instantiations. Then for every $L \geq 2$,*

$$\|\Delta_L\|_2 \leq \frac{t}{\beta c_{\Psi}} \left(\Delta_{\text{out}} L + \Delta_X \sqrt{L} \right) + \frac{C_{\text{sh}}}{c_{\Psi}} L z_{2\alpha}(\beta), \quad (23)$$

where

$$z_{2\alpha}(\beta) := \int_G e^{-2\alpha\beta A_{\pi}(g)} dg. \quad (24)$$

Proof. Lemma 5 gives $\|\Delta_L\|_2 \leq \epsilon_L^{(C)} + (t\Delta_X/(\beta\sqrt{2}c_\Psi))\sqrt{m_L}$. In the midline strip geometry, $m_L \leq 2L$, hence $\frac{1}{\sqrt{2}}\sqrt{m_L} \leq \sqrt{L}$. The explicit midline surrogate error bound is $\epsilon_L^{(C)} \leq (t\Delta_{\text{out}}/(\beta c_\Psi))L + (C_{\text{sh}}/c_\Psi)Lz_{2\alpha}(\beta)$ (from the outside-factor Efron–Stein control and Corollary 1). Insert these bounds to obtain (23). $\square$

4.6 Dyadic scales and a pivot schedule

We specialize to dyadic $L = 2^n$ and the pivot schedule

$$\alpha(L) := 1 - \frac{\kappa}{|P_\perp(\mathbb{B}_L)|}, \quad |P_\perp(\mathbb{B}_L)| = 8L(L+1), \quad (25)$$

so that $t/\beta = \kappa/(8L(L+1))$. Write $\alpha_n := \alpha(2^n)$ and $\Delta_n := \Delta_{2^n}$.

Corollary 2 (Explicit n -dependence of $\|\Delta_n\|_2$). *For every $n \geq 1$,*

$$\|\Delta_n\|_2 \leq \frac{\kappa\Delta_{\text{out}}}{8c_\Psi} \frac{1}{2^n+1} + \frac{\kappa\Delta_X}{8c_\Psi} \frac{1}{2^{n/2}(2^n+1)} + \frac{C_{\text{sh}}}{c_\Psi} 2^n z_{2\alpha_n}(\beta). \quad (26)$$

In particular,

$$\|\Delta_n\|_2 \leq \frac{\kappa\Delta_{\text{out}}}{8c_\Psi} 2^{-n} + \frac{\kappa\Delta_X}{8c_\Psi} 2^{-3n/2} + \frac{C_{\text{sh}}}{c_\Psi} 2^n z_{2\alpha_n}(\beta). \quad (27)$$

Proof. Insert $t/\beta = \kappa/(8L(L+1))$ into (23) and set $L = 2^n$. The simplified inequalities use $1/(2^n+1) \leq 2^{-n}$ and $1/(2^{n/2}(2^n+1)) \leq 2^{-3n/2}$. $\square$

4.7 Weak-coupling bound for $z_{2\alpha}(\beta)$

Lemma 9 (Explicit bound for $z_{2\alpha}(\beta)$). *Let $d_G := \dim(G)$. Assume the one-plaquette action A_π has a unique minimum at e and admits constants $m_- > 0$, $J_{\max} < \infty$, $a_0 > 0$ and a chart $g = \exp X$ on a neighborhood $\mathcal{U}_0$ of e with radius $r_0 > 0$ such that*

$$A_\pi(\exp X) \geq \frac{m_-}{2} \|X\|^2, \quad dg = J(X) dX, \quad 0 \leq J(X) \leq J_{\max} \quad (\|X\| \leq r_0),$$

and $A_\pi(g) \geq a_0$ on $G \setminus \mathcal{U}_0$. Then for every $\beta > 0$ and $\alpha \in (0, 1]$,

$$z_{2\alpha}(\beta) \leq J_{\max} \left(\frac{\pi}{\alpha\beta m_-} \right)^{d_G/2} + e^{-2\alpha\beta a_0}. \quad (28)$$

Proof. Split the integral over $\mathcal{U}_0$ and its complement. On $\mathcal{U}_0$ write $g = \exp X$ and use the quadratic lower bound on A_π and $J(X) \leq J_{\max}$:

$$\int_{\mathcal{U}_0} e^{-2\alpha\beta A_\pi(g)} dg \leq J_{\max} \int_{\|X\| \leq r_0} e^{-\alpha\beta m_- \|X\|^2} dX \leq J_{\max} \int_{\mathbb{R}^{d_G}} e^{-\alpha\beta m_- \|X\|^2} dX = J_{\max} \left(\frac{\pi}{\alpha\beta m_-} \right)^{d_G/2}.$$

On $G \setminus \mathcal{U}_0$ use $A_\pi(g) \geq a_0$ and $\int dg = 1$ to get $\int_{G \setminus \mathcal{U}_0} e^{-2\alpha\beta A_\pi(g)} dg \leq e^{-2\alpha\beta a_0}$. Sum the contributions. $\square$

5 Closing the finite-depth transfer budget: explicit $\kappa(\delta)$ and $\beta_0(\delta)$

We now choose κ (pivot slack) and a weak-coupling threshold β_0 so that

$$\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \delta. \quad (29)$$

5.1 Summability up to the mixing depth

Proposition 3 (Summable transfer budget). *Fix $\delta \in (0, 1)$ and let $k_{\text{opt}}(\beta)$ be the optimal mixing depth from Proposition 1 for block factor $L = 2$. Set $m_{\text{opt}}(\beta) = \max\{1, \beta s_\delta\}$. Assume the pivot schedule (25) and define $\alpha_{\min} := 1 - \kappa/48$. Let*

$$Z_\star(\beta) := J_{\max} \left(\frac{\pi}{\alpha_{\min} \beta m_-} \right)^{d_G/2} + e^{-2\alpha_{\min} \beta a_0}.$$

Then for all $\beta > 0$,

$$\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \frac{\kappa}{8c_\Psi} \left(2\Delta_{\text{out}} + \frac{\Delta_X}{1 - 2^{-3/2}} \right) + \frac{2C_{\text{sh}}}{c_\Psi} \sqrt{m_{\text{opt}}(\beta)} Z_\star(\beta). \quad (30)$$

Proof. Sum the dyadic bound (26) over $n < k_{\text{opt}}(\beta)$. For the first two sums use

$$\sum_{n \geq 1} \frac{1}{2^n + 1} \leq \sum_{n \geq 1} 2^{-n} = 1, \quad \sum_{n \geq 1} \frac{1}{2^{n/2}(2^n + 1)} \leq \sum_{n \geq 1} 2^{-3n/2} = \frac{2^{-3/2}}{1 - 2^{-3/2}} \leq \frac{1}{1 - 2^{-3/2}}.$$

For the shared-edge term, $\alpha_n \geq \alpha_{\min}$ so Lemma 9 gives $z_{2\alpha_n}(\beta) \leq Z_\star(\beta)$ uniformly in n . Therefore

$$\sum_{n < k_{\text{opt}}(\beta)} 2^n z_{2\alpha_n}(\beta) \leq \left(\sum_{n < k_{\text{opt}}(\beta)} 2^n \right) Z_\star(\beta) \leq 2^{k_{\text{opt}}(\beta)} Z_\star(\beta).$$

Since $4^{k_{\text{opt}}(\beta)} \geq m_{\text{opt}}(\beta)$, we have $2^{k_{\text{opt}}(\beta)} \leq 2\sqrt{m_{\text{opt}}(\beta)}$. Insert into the sum. $\square$

5.2 An explicit polynomial envelope and the final parameter choice

Lemma 10 (Elementary bound $e^{-x} \leq (p/e)^p x^{-p}$). *Let $p > 0$. Then for all $x > 0$, $e^{-x} \leq (p/e)^p x^{-p}$.*

Proof. Maximize $x^p e^{-x}$; the maximum occurs at $x = p$ with value $p^p e^{-p}$. $\square$

Corollary 3 (Polynomial envelope for $Z_\star(\beta)$). *Let $p = d_G/2$. Then*

$$Z_\star(\beta) \leq C_Z(\kappa) \beta^{-d_G/2}, \quad (31)$$

where

$$C_Z(\kappa) := J_{\max} \left(\frac{\pi}{\alpha_{\min} m_-} \right)^{d_G/2} + \left(\frac{d_G}{2e} \right)^{d_G/2} (2\alpha_{\min} a_0)^{-d_G/2}, \quad \alpha_{\min} = 1 - \kappa/48.$$

Proof. The first term of $Z_\star$ is already $\propto \beta^{-d_G/2}$. For the second, apply Lemma 10 with $x = 2\alpha_{\min} \beta a_0$ and $p = d_G/2$. $\square$

Proposition 4 (Explicit κ_δ and $\beta_0(\delta)$ forcing (29)). *Fix $\delta \in (0, 1)$. Define*

$$A_{\text{out}} := 2\Delta_{\text{out}} + \frac{\Delta_X}{1 - 2^{-3/2}}.$$

Set

$$\kappa_\delta := \min \left\{ 24, \frac{4c_\Psi}{A_{\text{out}}} \delta \right\}, \quad \alpha_{\min} := 1 - \kappa_\delta/48 \geq \frac{1}{2}, \quad (32)$$

and let $C_Z := C_Z(\kappa_\delta)$ be as in Corollary 3. Define

$$\beta_0(\delta) := \max \left\{ \frac{1}{s_\delta}, 1, \left(\frac{4C_{\text{sh}}}{c_\Psi} \sqrt{s_\delta} C_Z \delta^{-1} \right)^{\frac{2}{d_G-1}} \right\}. \quad (33)$$

Then for all $\beta \geq \beta_0(\delta)$, under the pivot schedule with $\kappa = \kappa_\delta$,

$$\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \delta. \quad (34)$$

Proof. Apply Proposition 3 with $\kappa = \kappa_\delta$. The first term becomes $\kappa_\delta A_{\text{out}}/(8c_\Psi) \leq \delta/2$ by (32). For the second term, if $\beta \geq 1/s_\delta$ then $m_{\text{opt}}(\beta) = \beta s_\delta$. Using Corollary 3,

$$\frac{2C_{\text{sh}}}{c_\Psi} \sqrt{m_{\text{opt}}(\beta)} Z_\star(\beta) \leq \frac{2C_{\text{sh}}}{c_\Psi} \sqrt{\beta s_\delta} C_Z \beta^{-d_G/2} = \frac{2C_{\text{sh}}}{c_\Psi} \sqrt{s_\delta} C_Z \beta^{-(d_G-1)/2}.$$

With $\beta \geq \beta_0(\delta)$, this is $\leq \delta/2$. Sum the two halves. $\square$

5.3 Summary of explicit constants and dependencies

For reference, we list the principal explicit quantities appearing in the entrance theorem.

- The mixing time s_δ is defined in Lemma 2 by $\|K_{s_\delta} - 1\|_2^2 = \delta$ and depends only on the compact group G .
- The blocking factor $B(\beta)$ is defined by (8) (with $L = 2$ in the applications below) and satisfies $B(\beta)^2/\beta \geq s_\delta$ at the target tolerance.
- The pivot slack κ_δ and the weak-coupling threshold $\beta_0(\delta)$ are given explicitly in Proposition 4. Their dependence is through the geometric constant C_{sh} , the group/representation parameters in Lemma 9, and the transfer constants $\Delta_{\text{out}}, \Delta_X, c_\Psi$ produced by the cut-local surrogate framework.
- The reference contraction constant q_* is as in Assumption 2(R1).
- The physical mass-gap lower bound m_* is as in Assumption 1(U3).

The constants C_{sh} and the local incidence bounds are purely combinatorial. The constants m_-, J_{max}, a_0 in Lemma 9 depend only on (G, π) via a fixed local chart near the identity.

6 From majorant mixing + transfer to Dobrushin entrance at block size $B(\beta)$

We now combine the mixing depth (majorant close to Haar) and the transfer budget (exact close to reference) into a Dobrushin entrance bound.

6.1 χ^2 control implies influence control

Lemma 11 (χ^2 control bounds total variation). *Let μ be a probability measure absolutely continuous with respect to Haar on G with density r . Then*

$$\|\mu - dg\|_{\text{TV}} \leq \frac{1}{2} \|r - 1\|_{L^1(dg)} \leq \frac{1}{2} \|r - 1\|_{L^2(dg)} = \frac{1}{2} \sqrt{D_{\chi^2}(\mu \| dg)}. \quad (35)$$

Proof. The first inequality is the definition of total variation. The second is Cauchy–Schwarz with $\int 1 dg = 1$. $\square$

Lemma 12 (A sufficient condition for Dobrushin contraction via near-Haar conditionals). *Let γ be a single-site specification on a graph such that each site x has an interaction neighborhood of size at most Δ (i.e. $\gamma_x(\cdot | \eta)$ depends on the boundary configuration η only through at most Δ sites). Assume there exists $\varepsilon_{\text{Haar}} \geq 0$ such that for all sites x and all boundary conditions η ,*

$$D_{\chi^2}(\gamma_x(\cdot | \eta) \| dg) \leq \varepsilon_{\text{Haar}}. \quad (36)$$

Then $\delta(\gamma) \leq \Delta \sqrt{\varepsilon_{\text{Haar}}}$. In particular, if $\Delta \sqrt{\varepsilon_{\text{Haar}}} < 1$ then γ is Dobrushin contractive.

Proof. By Lemma 11, uniformly in x, η one has $\|\gamma_x(\cdot | \eta) - dg\|_{\text{TV}} \leq \frac{1}{2}\sqrt{\varepsilon_{\text{Haar}}}$. Therefore for boundary conditions η, η' differing at a single site,

$$\|\gamma_x(\cdot | \eta) - \gamma_x(\cdot | \eta')\|_{\text{TV}} \leq \|\gamma_x(\cdot | \eta) - dg\|_{\text{TV}} + \|dg - \gamma_x(\cdot | \eta')\|_{\text{TV}} \leq \sqrt{\varepsilon_{\text{Haar}}}.$$

Summing influence coefficients over at most Δ sites in the interaction neighborhood yields $\delta(\gamma) \leq \Delta\sqrt{\varepsilon_{\text{Haar}}}$. $\square$

Proposition 5 (Dobrushin comparison from a conditional χ^2 gate). *Let $\gamma^{(\alpha)}$ be a single-site specification with Dobrushin constant $\delta(\gamma^{(\alpha)}) \leq q_* < 1$. Let γ^{ex} be another specification on the same single-site space. Assume that for every site x and every boundary condition η ,*

$$D_{\chi^2}(\gamma_x^{\text{ex}}(\cdot | \eta) \| \gamma_x^{(\alpha)}(\cdot | \eta)) \leq \varepsilon, \quad (37)$$

uniformly. Then

$$\delta(\gamma^{\text{ex}}) \leq q_* + \sqrt{\varepsilon}. \quad (38)$$

Proof. For fixed x and boundary conditions η, η' differing only at one site y , write

$$\|\gamma_x^{\text{ex}}(\cdot | \eta) - \gamma_x^{\text{ex}}(\cdot | \eta')\|_{\text{TV}} \leq \|\gamma_x^{\text{ex}}(\cdot | \eta) - \gamma_x^{(\alpha)}(\cdot | \eta)\|_{\text{TV}} + \|\gamma_x^{(\alpha)}(\cdot | \eta) - \gamma_x^{(\alpha)}(\cdot | \eta')\|_{\text{TV}} + \|\gamma_x^{(\alpha)}(\cdot | \eta') - \gamma_x^{\text{ex}}(\cdot | \eta')\|_{\text{TV}}.$$

The middle term is bounded by the Dobrushin influence coefficient of $\gamma^{(\alpha)}$. For the first and third term, use Lemma 11 with the reference measure $\gamma_x^{(\alpha)}(\cdot | \eta)$ and the χ^2 bound (37) to get $\text{TV} \leq \frac{1}{2}\sqrt{\varepsilon}$. Thus each influence coefficient of γ^{ex} is bounded by the corresponding coefficient of $\gamma^{(\alpha)}$ plus $\sqrt{\varepsilon}$, whence $\delta(\gamma^{\text{ex}}) \leq \delta(\gamma^{(\alpha)}) + \sqrt{\varepsilon} \leq q_* + \sqrt{\varepsilon}$. $\square$

6.2 One-shot entrance theorem (finite depth, no stable manifold)

Lemma 13 (Scale-wise product representation and the blocking filtration). *Let ν be a product probability measure $\nu = \nu_1 \otimes \cdots \otimes \nu_k$ on a product space $X = X_1 \times \cdots \times X_k$, and let $\mathcal{F}_n$ be the σ -algebra generated by the first n coordinate blocks. If $r_n \geq 0$ depends only on the n -th coordinate block and satisfies $\mathbb{E}_{\nu_n}[r_n] = 1$, then*

$$\mathbb{E}_{\nu}[r_n | \mathcal{F}_{n-1}] = 1, \quad \mathbb{E}_{\nu}[(r_n - 1)^2 | \mathcal{F}_{n-1}] = \mathbb{E}_{\nu_n}[(r_n - 1)^2]$$

almost surely. In particular, if $\|r_n - 1\|_{L^2(\nu)} = \|r_n - 1\|_{L^2(\nu_n)} \leq \eta_n$, then the hypotheses of Lemma 14 hold with the deterministic bounds η_n .

Proof. Since ν factorizes, conditioning on $\mathcal{F}_{n-1}$ leaves the n -th coordinate distributed according to ν_n and independent of $\mathcal{F}_{n-1}$. Therefore $\mathbb{E}_{\nu}[r_n | \mathcal{F}_{n-1}] = \mathbb{E}_{\nu_n}[r_n] = 1$. The same factorization gives $\mathbb{E}_{\nu}[(r_n - 1)^2 | \mathcal{F}_{n-1}] = \mathbb{E}_{\nu_n}[(r_n - 1)^2]$. $\square$

Lemma 14 (Multiplicative χ^2 accumulation). *Let $(\mathcal{F}_j)_{j=0}^k$ be an increasing filtration on a probability space $(\Omega, \mathcal{F}, \nu)$. Let $r_j \geq 0$ be $\mathcal{F}_j$ -measurable random variables such that*

$$\mathbb{E}_{\nu}[r_j | \mathcal{F}_{j-1}] = 1 \quad \text{a.s.}$$

and suppose there exist deterministic numbers $\eta_j \geq 0$ such that

$$\mathbb{E}_{\nu}[(r_j - 1)^2 | \mathcal{F}_{j-1}] \leq \eta_j^2 \quad \text{a.s.}$$

Define $R_k := \prod_{j=1}^k r_j$. Then $\mathbb{E}_{\nu}[R_k] = 1$ and

$$D_{\chi^2}(R_k \cdot \nu | \nu) = \mathbb{E}_{\nu}[(R_k - 1)^2] = \mathbb{E}_{\nu}[R_k^2] - 1 \leq \prod_{j=1}^k (1 + \eta_j^2) - 1 \leq \exp\left(\sum_{j=1}^k \eta_j^2\right) - 1.$$

Proof. Since $\mathbb{E}_\nu[r_j \mid \mathcal{F}_{j-1}] = 1$, the process R_j is a nonnegative martingale and $\mathbb{E}_\nu[R_k] = 1$. Moreover,

$$\mathbb{E}_\nu[R_j^2 \mid \mathcal{F}_{j-1}] = R_{j-1}^2 \mathbb{E}_\nu[r_j^2 \mid \mathcal{F}_{j-1}] = R_{j-1}^2 \left(1 + \mathbb{E}_\nu[(r_j - 1)^2 \mid \mathcal{F}_{j-1}]\right) \leq R_{j-1}^2(1 + \eta_j^2).$$

Taking expectations and iterating gives $\mathbb{E}_\nu[R_k^2] \leq \prod_{j=1}^k (1 + \eta_j^2)$. Finally, $D_{\chi^2}(R_k \cdot \nu \parallel \nu) = \mathbb{E}_\nu[(R_k - 1)^2] = \mathbb{E}_\nu[R_k^2] - 1$ because $\mathbb{E}_\nu[R_k] = 1$. The exponential bound uses $1 + x \leq e^x$. $\square$

Theorem 2 (Finite-depth entrance at $B(\beta)$). *Fix $\delta \in (0, 1)$ and set $\underline{\delta} := \delta/2$. Choose κ_δ and $\beta_0(\underline{\delta})$ as in Proposition 4. Let $B(\beta) = 2^{k_{\text{opt}}(\beta)}$ be the majorant mixing scale from (8) with $L = 2$ and target tolerance $\underline{\delta}$ (i.e. with s_δ in (8)). Then for every $\beta \geq \beta_0(\underline{\delta})$:*

1. (Majorant mixing) $D_{\chi^2}(\rho_\beta^{(*B(\beta)^2)} \parallel dg) \leq \underline{\delta}$;

2. (Transfer budget) $\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2 \leq \underline{\delta}$.

Moreover, at block size $B(\beta)$ the exact blocked specification is in a Dobrushin contraction basin $\delta(\gamma^{\text{ex}}) \leq q_* + \delta$, where $q_* < 1$ is the Dobrushin constant of the reference specification. In particular, if $q_* + \delta < 1$, then γ^{ex} has a unique infinite-volume Gibbs state at that block size and exponential clustering holds.

Proof. Item (1) is Proposition 1 with $L = 2$ and $m = B(\beta)^2$, applied with target tolerance $\underline{\delta}$. Item (2) is Proposition 4, applied with tolerance $\underline{\delta}$.

The finite-depth transfer mechanism expresses the exact-to-reference comparison at block size $B(\beta)$ as a product of per-step normalized tilts $r_n = 1 + \Delta_n$ along a blocking filtration $(\mathcal{F}_n)$, with conditional normalization $\mathbb{E}[r_n \mid \mathcal{F}_{n-1}] = 1$. In the scale-wise product representation of the reference measure used to bound the per-step defects (cf. Lemma 13), each r_n depends only on the newly revealed scale- n coordinates, and therefore

$$\mathbb{E}[(r_n - 1)^2 \mid \mathcal{F}_{n-1}] = \mathbb{E}[(r_n - 1)^2] = \|\Delta_n\|_2^2$$

a.s. Lemma 14 therefore yields the conditional χ^2 gate (37) with

$$\varepsilon \leq \exp\left(\sum_{n < k_{\text{opt}}(\beta)} \|\Delta_n\|_2^2\right) - 1 \leq \exp(\underline{\delta}^2) - 1.$$

Since $\underline{\delta} \leq 1$, we have $\exp(\underline{\delta}^2) - 1 \leq (e - 1)\underline{\delta}^2$, and hence $\sqrt{\varepsilon} \leq \sqrt{e - 1} \underline{\delta} = \frac{\sqrt{e - 1}}{2} \delta < \delta$. Applying Proposition 5 yields $\delta(\gamma^{\text{ex}}) \leq \delta(\gamma^{(\alpha)}) + \sqrt{\varepsilon} \leq q_* + \delta$.

If $q_* + \delta < 1$, uniqueness and exponential clustering follow from the standard Dobrushin theorem. $\square$

Remark 8 (Reference contraction constant q_*). *The constant $q_* < 1$ is as in Assumption 2(R1). The finite-depth entrance argument uses only the uniform bound $\delta(\gamma^{(\alpha)}) \leq q_*$ for the chosen reference specification in finite volume and under admissible boundary conditions.*

7 Exponential clustering and the physical mass gap

7.1 Dobrushin contraction implies exponential decay

Lemma 15 (Exponential decay from Dobrushin). *Let γ be a specification on a graph with Dobrushin constant $\delta(\gamma) \leq q < 1$. Then for bounded local functions F, G supported on sets S, T ,*

$$|\langle FG \rangle_c| \leq \frac{1}{1 - q} \|F\|_\infty \|G\|_\infty q^{\text{dist}(S, T)} = \frac{1}{1 - q} \|F\|_\infty \|G\|_\infty e^{-(\log q) \text{dist}(S, T)}. \quad (39)$$

Proof. This is the standard Dobrushin mixing estimate: the influence matrix bounds covariances by a sum over influence paths, which is a geometric series controlled by q . The explicit constant and rate follow from bounding the path sum by $\sum_{n \geq \text{dist}(S, T)} q^n = q^{\text{dist}(S, T)} / (1 - q)$. $\square$

7.2 OS spectral gap extraction from Euclidean decay

We now spell out the OS spectral-gap extraction in a form that makes constants explicit.

Lemma 16 (OS semigroup representation). *Let $\langle \cdot \rangle$ be an OS-positive Schwinger functional on a translation-invariant algebra $\mathcal{A}$ with positive-time subalgebra $\mathcal{A}_+$ and reflection Θ . Let $(\mathcal{H}, \Omega, H)$ be the OS reconstruction. Then for every $F \in \mathcal{A}_+$ and $t \geq 0$,*

$$\langle F \Theta \tau_t F \rangle = \langle [F], e^{-tH} [F] \rangle_{\mathcal{H}}, \quad (40)$$

and $[F - \langle F \rangle] \perp \Omega$.

Proof. This is standard OS reconstruction: the inner product is $(F, G) = \langle F \Theta G \rangle$ on $\mathcal{A}_+$ modulo nulls. Time translations act by a symmetric contraction semigroup $T(t)$, hence $T(t) = e^{-tH}$. $\square$

Lemma 17 (Laplace-transform gap criterion). *Let $H \geq 0$ be self-adjoint and $\psi \in \mathcal{H}$. If there exist $m > 0$ and $C < \infty$ such that*

$$\langle \psi, e^{-tH} \psi \rangle \leq C e^{-mt} \quad \forall t \geq 0, \quad (41)$$

then the spectral measure of H associated to ψ satisfies $\mu_\psi([0, m)) = 0$.

Proof. By the spectral theorem, $\langle \psi, e^{-tH} \psi \rangle = \int_0^\infty e^{-t\lambda} d\mu_\psi(\lambda)$. Fix $\varepsilon \in (0, m)$. Then $\int_{[0, m-\varepsilon]} e^{-t\lambda} d\mu_\psi \geq e^{-t(m-\varepsilon)} \mu_\psi([0, m-\varepsilon])$. Combine with (41) to obtain $\mu_\psi([0, m-\varepsilon]) \leq C e^{-\varepsilon t}$ for all t , hence the measure is 0. Let $\varepsilon \downarrow 0$. $\square$

Theorem 3 (Spectral gap from Euclidean exponential decay). *Let $\langle \cdot \rangle$ be an OS-positive Schwinger functional with OS reconstruction $(\mathcal{H}, \Omega, H)$. Assume that for every $F \in \mathcal{A}_+$,*

$$\langle (F - \langle F \rangle) \Theta \tau_t (F - \langle F \rangle) \rangle \leq \langle (F - \langle F \rangle) \Theta (F - \langle F \rangle) \rangle e^{-mt} \quad \forall t \geq 0, \quad (42)$$

for some $m > 0$. Then $\text{Spec}(H) \subset \{0\} \cup [m, \infty)$.

Proof. Fix $F \in \mathcal{A}_+$ and set $F_0 := F - \langle F \rangle$. Lemma 16 gives $\langle F_0 \Theta \tau_t F_0 \rangle = \langle [F_0], e^{-tH} [F_0] \rangle$ and $[F_0] \perp \Omega$. Then (42) implies $\langle [F_0], e^{-tH} [F_0] \rangle \leq \| [F_0] \|^2 e^{-mt}$. Apply Lemma 17 to $\psi = [F_0]$. Since the orthogonal complement of Ω is spanned by such vectors, the spectral inclusion follows. $\square$

8 Scope and status of the result

Observable algebra and locality

The continuum limit constructed here is formulated on a local net of gauge-invariant, bounded smeared observables at fixed macroscopic scale $\ell > 0$, denoted $\mathcal{O}_\ell$. Observables are localized in bounded regions in the usual Euclidean sense inherited from the lattice support of their defining functionals and the blocking maps. The limiting Schwinger functional on $\mathcal{O}_\ell$ is translation invariant and Osterwalder–Schrader positive. By OS reconstruction this yields a Hilbert space representation with a positivity-preserving semigroup and the corresponding Minkowski-space reconstruction on the reconstructed observable algebra.

Mass gap statement

The mass gap is extracted from exponential decay of Euclidean correlations (obtained from uniform Dobrushin contraction at the blocking scale) via the OS semigroup representation and a Laplace-transform criterion. The conclusion is a strictly positive lower bound on the spectrum of the reconstructed Hamiltonian above the vacuum, as stated in Theorem 1.

What is not addressed

This manuscript does not construct sharp local operator-valued distributions obtained by taking a $t \rightarrow 0$ limit, nor does it address the existence of unsmeared local fields at arbitrarily small scales. The present results are instead established on the bounded macroscopic algebra $\mathcal{O}_\ell$ at fixed positive scale ℓ . This limitation does not affect the spectral-gap conclusion on $\mathcal{O}_\ell$, since the gap follows from Euclidean exponential clustering and OS reconstruction on that algebra.

External UV inputs

The continuum statement is conditional on explicit ultraviolet tuning and identification requirements (Assumption 1), which isolate the conversion from a finite-depth entrance estimate on the blocked lattice specification to a continuum limit at fixed macroscopic scale. The finite-depth entrance mechanism and the explicit constants $(B(\beta), \kappa_\delta, \beta_0(\delta))$ are proved internally in this manuscript.